A Few Useful Tricks

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1 Trick 1: $\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+1}$

This incredibly useful bit of mathematical magic works when the denominators of the two fractions are of the form n and $n+1$, their numerators are both 1, and $n > 0$. For example, something like

$$\left(\frac{1}{3}\right)\left(\frac{1}{4}\right) = \frac{1}{12} \text{ and } \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}$$

So why does this work? Consider

$$\begin{aligned}
\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) &= \frac{1}{n \cdot (n+1)} && \text{\# numerators are both 1} \\
&= \frac{(n-n)+1}{n \cdot (n+1)} && \text{\# } (n-n) = 0 \text{ and } 0+1 = 1 \\
&= \frac{(n+1)-n}{n \cdot (n+1)} && \text{\# rearrange (+ is commutative and associative)} \\
&= \frac{(n+1)}{n \cdot (n+1)} - \frac{n}{n \cdot (n+1)} && \text{\# split into a difference} \\
&= \frac{1}{n} - \frac{n}{n \cdot (n+1)} && \text{\# cancel } (n+1) \text{ in the first term} \\
&= \frac{1}{n} - \frac{1}{n+1} && \text{\# cancel } n \text{ in the second term } \square
\end{aligned}$$

So we have

$$\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+1} \tag{1}$$

An example of the use of this trick in evaluating infinite series is Burkard Polster's (aka Mathologer) use of Equation 1 when describing how Euler's solution to the Basel problem works [1].

1.1 Telescoping Series

What is a telescoping series? Briefly, a telescoping series is a sum that is characterized by the fact that the partial sums of the series (called telescoping sums) contain pairs of consecutive terms which cancel each other, leaving only the first and final terms. This cancellation of adjacent terms is whimsically called "collapsing the telescope".

For example, consider the infinite series whose terms are described by Equation 1. This is a telescoping series, which can be seen in the following

$$\begin{aligned}
\sum_{n=1}^{\infty}\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) &= \sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right) && \text{\# Equation 1 (this is trick 1)}\\
&= \lim_{N\to\infty}\sum_{n=1}^{N}\left(\frac{1}{n}-\frac{1}{n+1}\right) && \text{\# we are interested in the limit}\\
&= \lim_{N\to\infty}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\cdots+\left(\frac{1}{N}-\frac{1}{N+1}\right)\right]\\
&= \lim_{N\to\infty}\left[1+\left(-\frac{1}{2}+\frac{1}{2}\right)+\left(-\frac{1}{3}+\frac{1}{3}\right)+\cdots+\left(-\frac{1}{N}+\frac{1}{N}\right)-\frac{1}{N+1}\right]\\
&= \lim_{N\to\infty}\left[1-\frac{1}{N+1}\right] && \text{\# collapse the telescope}\\
&= 1
\end{aligned}$$

A bit more formally, a telescoping series is the sum of terms of the form $(a_j - a_{j+1})$. As a result of the structure of its terms a telescoping series has the following property:

$$\begin{aligned}
\sum_{j=1}^{n}(a_j - a_{j+1}) &= (a_1-a_2)+(a_2-a_3)+(a_3-a_4)+\cdots+(a_n-a_{n+1}) && \text{\# expand terms}\\
&= a_1+(a_2-a_2)+(a_3-a_3)+\cdots+(a_n-a_n)-a_{n+1} && \text{\# group like terms}\\
&= a_1-a_{n+1} && \text{\# simplify}
\end{aligned}$$

1.2 Triangular Numbers

The example I want to explore here is the infinite sum of the reciprocals of the triangular numbers [4]. So first, what is a triangular number? The first six triangular numbers are depicted as triangles in Figure 1. We can see from the figure that a triangular number T_n is the sum of the integers from 1 to n.

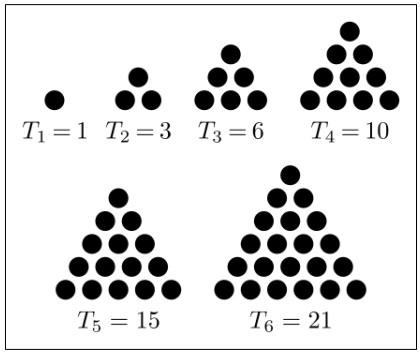

Figure 1: The First Six Triangular Numbers [4]

Since T_n is just the sum of the integers from 1 to n we can easily calculate the first few values of T_n, which are 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105 and so on (note that some authors include 0 in the triangular number sequence).

Ok, but what is the general form of T_n? Since T_n is the sum of the first n integers we know that

$$T_n = \sum_{k=1}^{n} k = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} = \binom{n+1}{2} \tag{2}$$

Interestingly if you start with the number 1 in row 2 of Pascal's Triangle [2] the corresponding diagonal contains the triangular numbers. This is shown in blue in Figure 2.

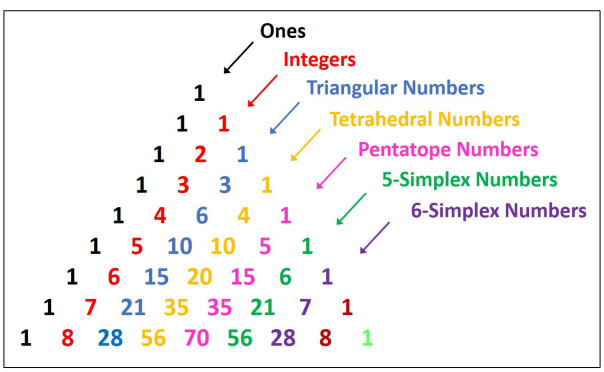

Figure 2: Number Sequences Found In Pascal's Triangle

As an aside, the German mathematician and scientist Carl Friedrich Gauss is said to have discovered the consecutive integer formula, $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$, as a young person; however, this history is at the very least controversial [4].

1.3 So What is the Sum of the Reciprocals of the Triangular Numbers?

All of this is good, but back to the question: what is the sum of the reciprocals of the triangular numbers? Well, first, the sum looks like

$$\begin{aligned}
\sum_{n=1}^{N} \frac{1}{T_n} &= \sum_{n=1}^{N} \frac{1}{\frac{n(n+1)}{2}} && \text{\# definition of } T_n \text{ (Equation 2)} \\
&= \sum_{n=1}^{N} \frac{2}{n(n+1)} && \text{\# simplify} \\
&= 2\sum_{n=1}^{N} \left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) && \text{\# pull 2 out of the sum and group terms} \\
&= 2\sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) && \text{\# Equation 1}
\end{aligned}$$

We saw in Section 1.1 that $\lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$ and we know from the above that the infinite sum of the reciprocals of the triangular numbers is twice this limit. So we know that

$$\sum_{n=1}^{\infty} \frac{1}{T_n} = 2\left[\lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right)\right] = 2 \cdot 1 = 2$$

1.4 A Slightly More General Form Of This Trick

A slightly more general form of this trick allows for denominators of the form n and $n+k$ rather than the n and $n+1$ denominators we saw in Section 1. The constraints on this form of the trick are that the denominators are of the form n and $n+k$, the product of the numerators must equal the absolute value of the difference of the denominators (that is, k), and $n > 0$.

If we consider the case in which $k = ab$ with $a \geq 0$, $b \geq 0$, and $n > 0$ we see that

$$
\begin{aligned}
\left(\frac{a}{n}\right)\left(\frac{b}{n+ab}\right) &= \frac{ab}{n\cdot(n+ab)} && \text{\# the product of the numerators } ab = |n-(n+ab)| \\
&= \frac{(n-n)+ab}{n\cdot(n+ab)} && \text{\# } (n-n)=0 \text{ and } 0+ab=ab \\
&= \frac{(n+ab)-n}{n\cdot(n+ab)} && \text{\# rearrange (+ is commutative and associative)} \\
&= \frac{(n+ab)}{n\cdot(n+ab)} - \frac{n}{n\cdot(n+ab)} && \text{\# split into a difference} \\
&= \frac{1}{n} - \frac{n}{n\cdot(n+ab)} && \text{\# cancel } (n+ab) \text{ in the first term} \\
&= \frac{1}{n} - \frac{1}{n+ab} && \text{\# cancel } n \text{ in the second term } \square
\end{aligned}
$$

So for $a \geq 0$, $b \geq 0$, and $n > 0$ we have

$$\left(\frac{a}{n}\right)\left(\frac{b}{n+ab}\right) = \frac{1}{n} - \frac{1}{n+ab} \tag{3}$$

To see how this works consider the following example: let $a = 2$, $b = 5$ and $n = 3$. Then

$$\left(\frac{2}{3}\right)\left(\frac{5}{3+(2\cdot 5)}\right) = \left(\frac{2}{3}\right)\left(\frac{5}{3+10}\right) = \left(\frac{2}{3}\right)\left(\frac{5}{13}\right) = \frac{10}{39}$$

and

$$\frac{1}{3} - \frac{1}{3+(2\cdot 5)} = \frac{1}{3} - \frac{1}{3+10} = \frac{1}{3} - \frac{1}{13} = \frac{13}{39} - \frac{3}{39} = \frac{13-3}{39} = \frac{10}{39}$$

We can also note that if $a = b = 1$ in Equation 3 we get the special case we considered in Section 1:

$$\left(\frac{1}{n}\right)\left(\frac{1}{n+(1\cdot 1)}\right) = \left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+(1\cdot 1)} = \frac{1}{n} - \frac{1}{n+1}$$

Equation 3 also works correctly for $a = 0$ and $b \neq 0$, $a \neq 0$ and $b = 0$, and $a = b = 0$.

Is this generalization useful? I don't know but it was fun trying to make it work. I'll just note here that if you sum terms of the "generalized" form it does not form a telescoping series, so that's a loss.

2 Conclusions

I'm not sure if Equation 3 is useful for anything but it was a fun exercise to do. On the other hand, the basic trick (Equation 1) turns out to be very useful, especially when evaluating infinite telescoping series.

References

[1] Burkard Polster (aka Mathologer). Euler's real identity NOT e to the i pi = -1. https://www.youtube.com/watch?v=yPl64xi_ZZA, Aug 2017. [Online; accessed 17-August-2021].

[2] Wikipedia contributors. Pascal's Triangle — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Pascal%27s_triangle&oldid=1036356492, 2021. [Online; accessed 18-August-2021].

[3] Wikipedia contributors. Telescoping Series — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Telescoping_series&oldid=1028468162, 2021. [Online; accessed 17-August-2021].

[4] Wikipedia contributors. Triangular Number — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Triangular_number&oldid=1034664425, 2021. [Online; accessed 17-August-2021].